

Prenatal diagnostic

Dr. Shayan Jalal Khalaf

FIBOG, MBChB

College of Medicine, Sulaimani University

Outline

- Introduction
- Indications of prenatal diagnosis
- First trimester combined test
- Triple test
- cffDNA
- Pre-Implantation diagnosis
- Chorion villus sampling
- Amniocentesis
- Cordocentesis
- Fetoskopie

Introduction

- Using a wide variety of screening and diagnostic tests to assess health of the fetus to:
 - Enable couples at risk of genetic disorders to have unaffected children
 - Identify potentially important intrauterine treatment if available
 - Plan for complications at birth
 - Decide to whether continue pregnancy
 - Discover conditions that may impact the future pregnancies

Indications for prenatal diagnosis

- Increased risk for chromosomal fetal abnormalities based on advanced maternal age, previous affected pregnancy
- Family history of a known genetic condition or congenital abnormalities
- Either of the couple affected by congenital disorder

Indications

- Previous history of a pregnancy affected by congenital anomalies
- Maternal medical conditions, medications, congenital infections
- Abnormal ultrasound findings- soft markers

Methods of prenatal diagnostics Non-Invasive and invasive

Non- invasive :

First trimester screening

Maternal serum biochemistry

NIPDT(Analysis of fetal cells in maternal blood)

Ultrasonography

Echocardiography

Doppler sonography

Preeclampsia screening

Pre-Implantation diagnosis

Methods of prenatal diagnostics

Non-Invasive and invasive

- **Invasive:**
- Chorion villus sampling
- Amniocentesis
- Cordocentesis
- Fetoskopie

First trimester combined test

- **A blood test :**

- 1 pregnancy-associated plasma protein-A (PAPP-A)
 - 2 human chorionic gonadotropin (HCG)
- PAPP-A levels are lower and free beta-hCG levels are higher in trisomy 21
 - PAPP-A levels and free beta-hCG levels are low in trisomy 18,13

- **An ultrasound exam:**

Nuchal translucency(NT)



frontomaxillary facial (FMF)

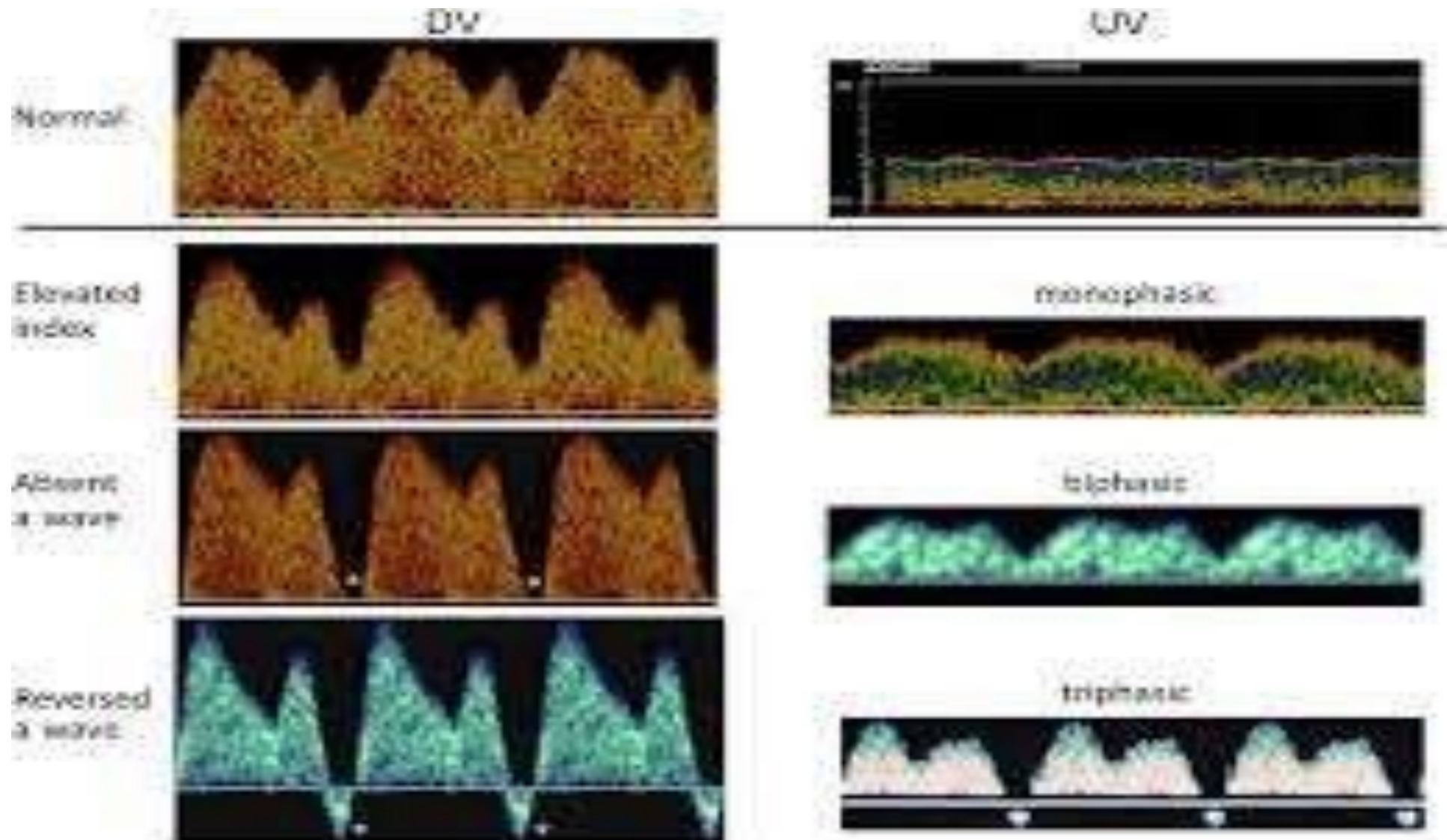


Present and Absent NB



Part 1
81
48
72

Assessment of ductus venosus flow at 11-13 weeks' gestation



Triple- Test

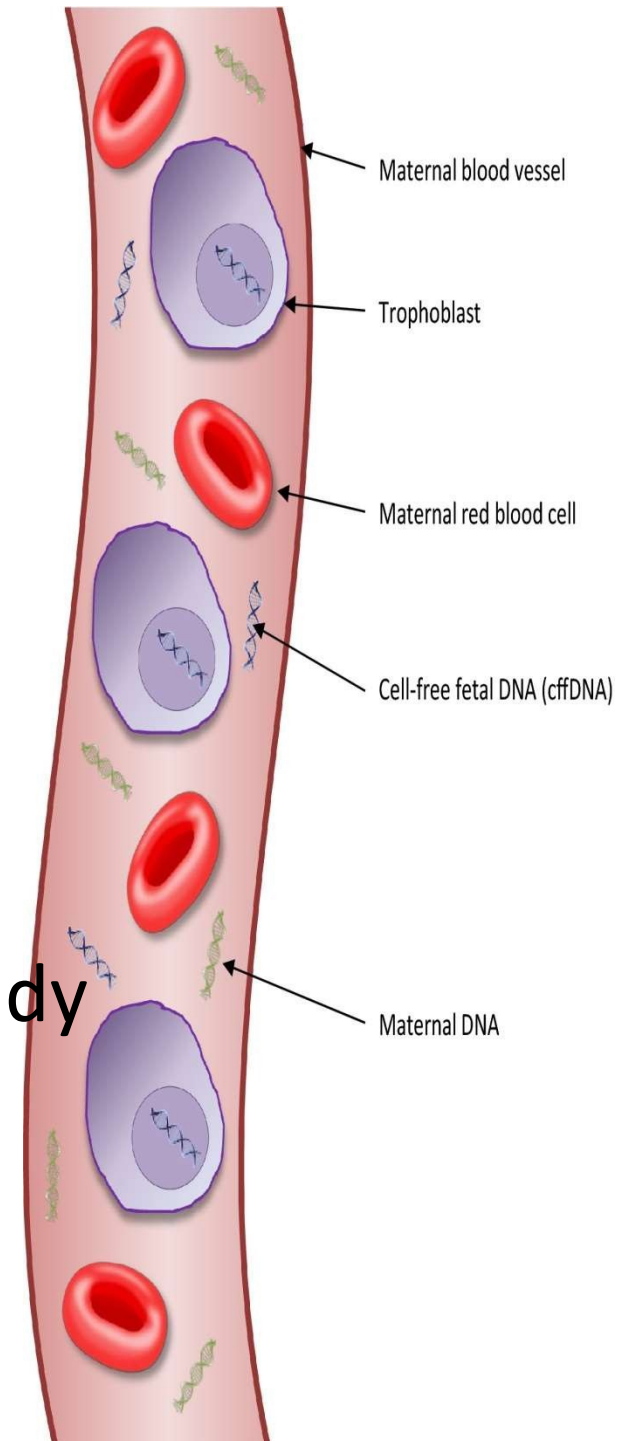
DR:58%, FPR:3%

The triple screening measures:

- 1 alpha-fetoprotein (AFP):** high in neural tube defects, abdominal wall defects,... and low in Down syndrome.
- 2 human chorionic gonadotropin (hCG)**
- 3 estriol (uE3):** low in Down syndrome or Edward's syndrome.

Cell free fetal DNA(cffDNA)

- Possible from week 9 + 0
- high sensitivity in trisomy 21 --->
- negative test ----> trisomy 21 very unlikely
- Indication:
 - a. Age>35 years old
 - b. sonographic evidence of aneuploidy
 - C- previous pregnancy with trisomy
 - d- positive triple test



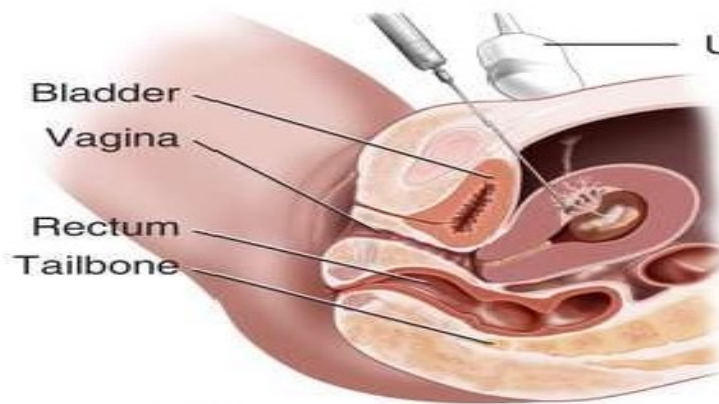
Preimplantation genetic Diagnosis(PGD)

- **What is Preimplantation Genetic Diagnosis?**
a reproductive technology during IVF cycle
- **Those who might benefit most from this test**
couples at increased risk for chromosome abnormalities or specific genetic disorders.

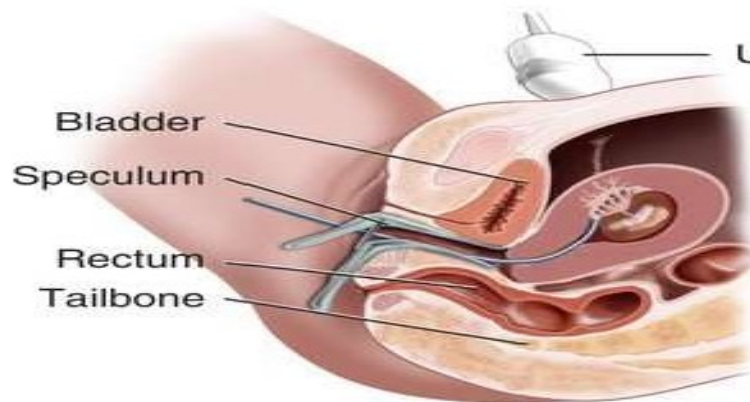
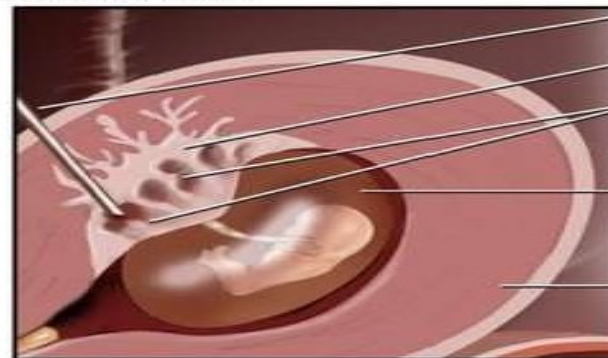
Chorionic villus sampling(CVS)

- **Chorionic villus sampling (CVS)** is a prenatal test in which a **sample** of **chorionic villi** is removed from the placenta for testing. The **sample** can be taken through the cervix (transcervical) or the abdominal wall (transabdominal).
- Implementation only after the 11 + 0 week of pregnancy, before 10 + 0 weeks → transverse fetal limb anomalies, microgenathy and microglossia
- miscarriage from CVS is approximately 2%.

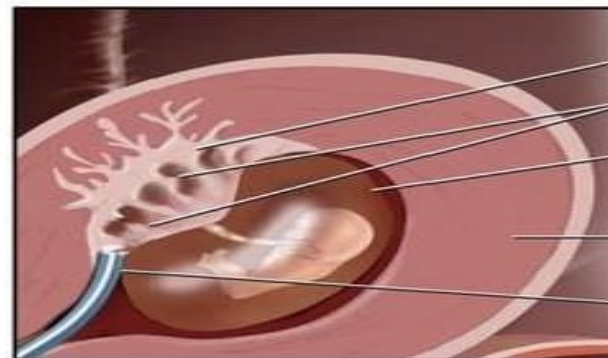
Chorionic Villus Sampling (CVS)



CVS Transabdominal



Transcervical CVS

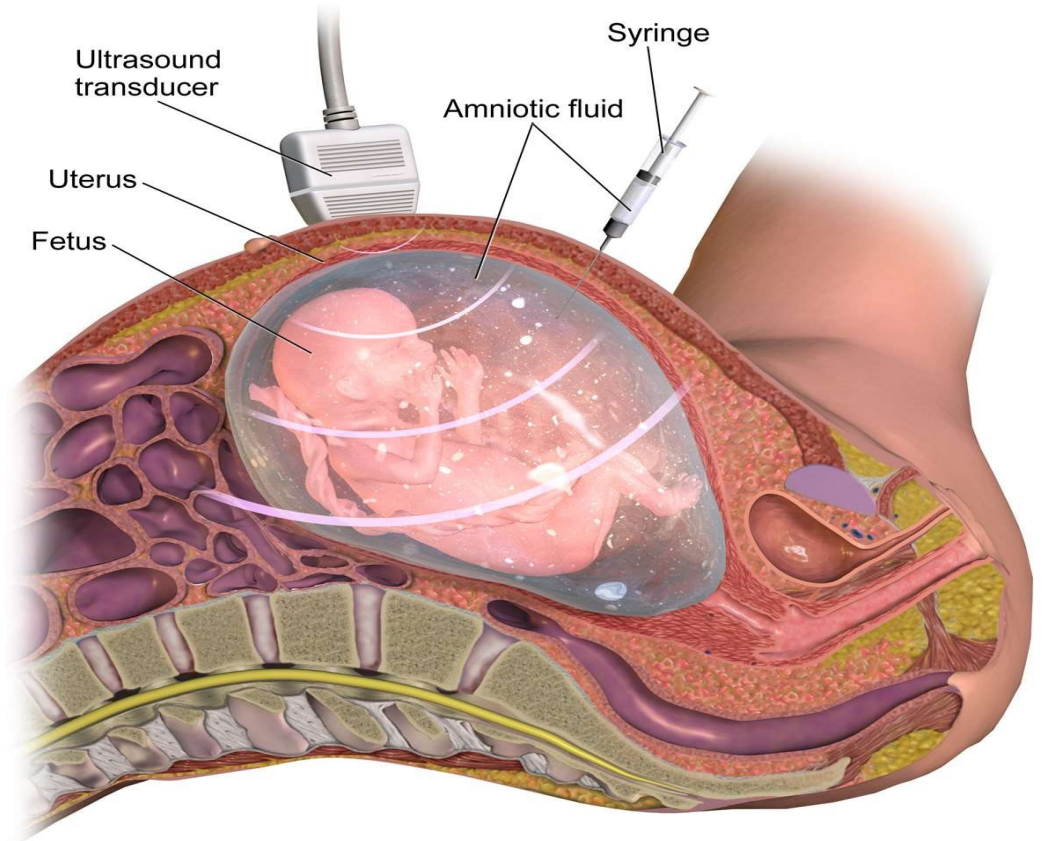


Chorionic villus sampling(CVS)

- Disadvantages:
- no AFP determination in amniotic fluid possible
- Mosaic formation is possible in the placenta and affects the information

Amniocentesis

- An amniocentesis procedure takes a sample (15–20 ml) of amniotic fluid that contains fetal cells.
- Abort risk 1,9%
- Implementation after the 15 + 0 weeks, before 15 + 0 weeks → increase Foot-hand deformities And abort risk.



Amniocentesis

- **Diagnostic:**
- Chromosomal analysis if the mother is >35 years old or positive family history or positive previous pregnancy
- Abnormal sonomorphologic finding
- To detect inherited metabolic disorders and other disorders such as muscular dystrophy.
- To check the levels of Alpha feto protein to rule out neural tube defects Such as spina bifid and abdominal wall defects such as omphalocele
- To detect fetal lung maturity
- For the measuring bilirubin levels to diagnose fetal haemolytic disease
- Measurement of AF-Insulin, if diabetic fetopathy is suspected
- Detection of toxoplasmosis or cytomegalovirus infection

Amniocentesis

- Therapeutic indications:
- For amnion reduction in case of polyhydramnion to
- For intrauterine transfusion



Chromosomic
Alterations



Genetic
Alterations



Fetal
Malformations

Cordocentesis

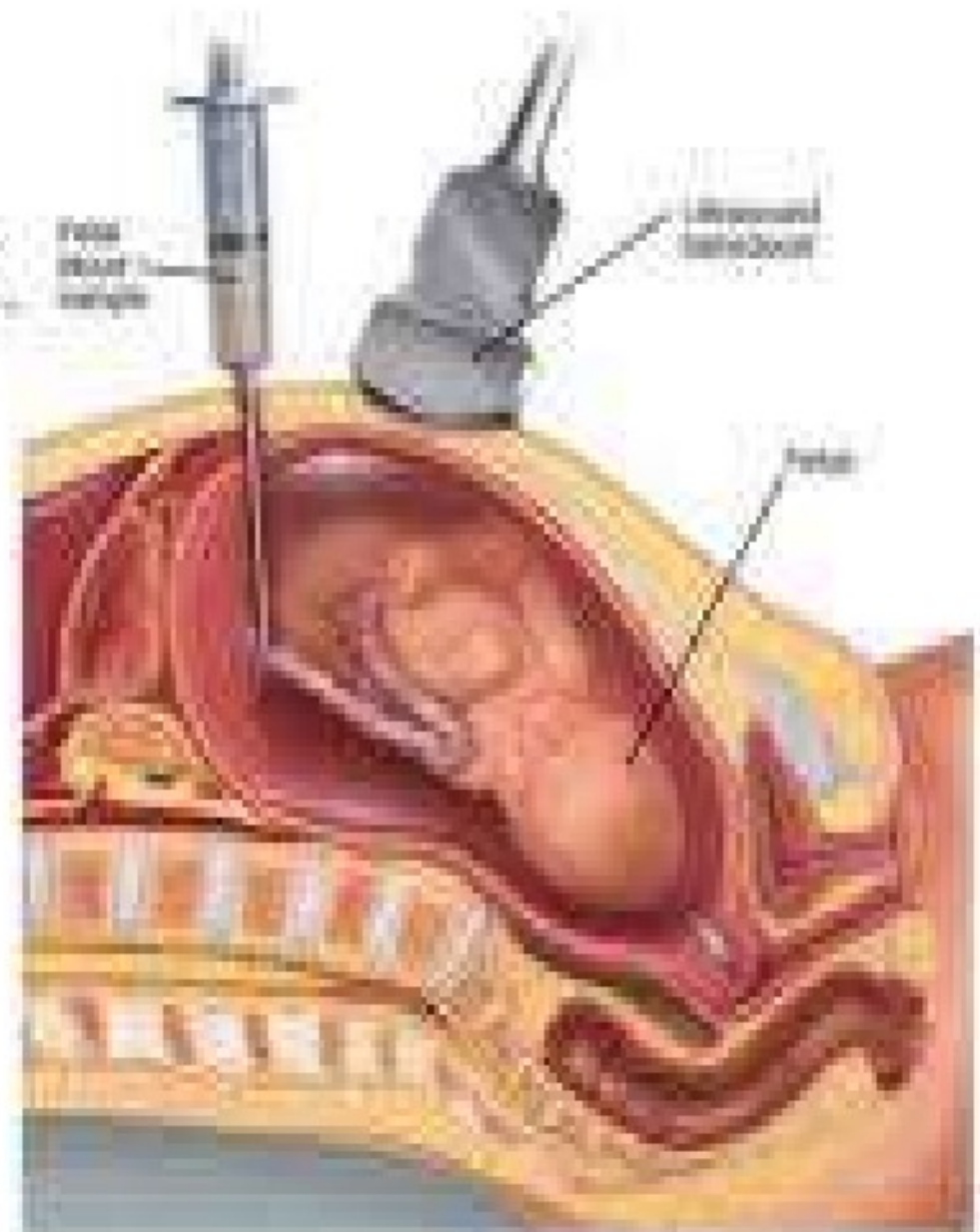
- Procedure:
 - Puncture of the umbilical vein, under ultrasound guidance, usually at its placental origin. Results typically take around 72 hours.
- Indications:
 - 1 Assessment and treatment of confirmed red cell or platelet alloimmunization.
 - 2 Assessment of congenital infections, if the mother has a positive IgM titer: Parvovirus B19, Toxoplasmosis, cytomegaly infection.
 - 3 Assessment of the fetal anemia
 - Parvovirus B19-infection und hemolysis in severe rhesus incompatibility
- 4- Unclear karyotyping from the amniotic fluid e.g. mosaic formation
- 5- Transfusion in severe fetal anemia (<10g%)
- 6- Therapy-refractory tachycardia of the fetus -> attempted therapy with amiodarone infused into the umbilical vein

Risks

- Fetal loss rate is 1%
- Bleeding from the umbilical cord usually stops after 10-20s
- Transient bradycardia (especially if the umbilical artery is punctured)
- By puncture the umbilical cord at fetal insertion, there is little risk of fetal injury.
- By puncture the umbilical cord at placental side , maternal and child blood can be mixed up

Cordocentesis may be performed to help diagnose any of the following concerns:

- Malformations of the fetus
- Fetal infection (i.e. toxoplasmosis or rubella)
- Fetal platelet count
- Fetal anemia
- Rh-Isimmunization



Fetoscopy

- Fetoscopy is an endoscopic procedure during pregnancy to allow access to the fetus, the amniotic cavity, the umbilical cord and the fetal side of the placenta. It is done during and after 18 weeks of pregnancy.

- **Procedure:**

A small (3–4 mm) incision is made in the abdomen, and an endoscope is inserted through the abdominal wall and uterus into the amniotic cavity.

- **Indications:**

- Clarification of suspected skin diseases:
- Epidermolysis and congenital Ichthyosis
- Fetoscopically controlled therapy: e.g. laser coagulation of the placental vascular anastomoses in fetofetal transfusion syndrome



CONCLUSIONS

- choose the right patient
- First trimester screening is optional.
- FTCT positive in intermediate risk patients → cffDNA
- FTCT positive in high risk patients → invasive PND
- positive NIPT → confirmatory diagnostic testing. invasive

